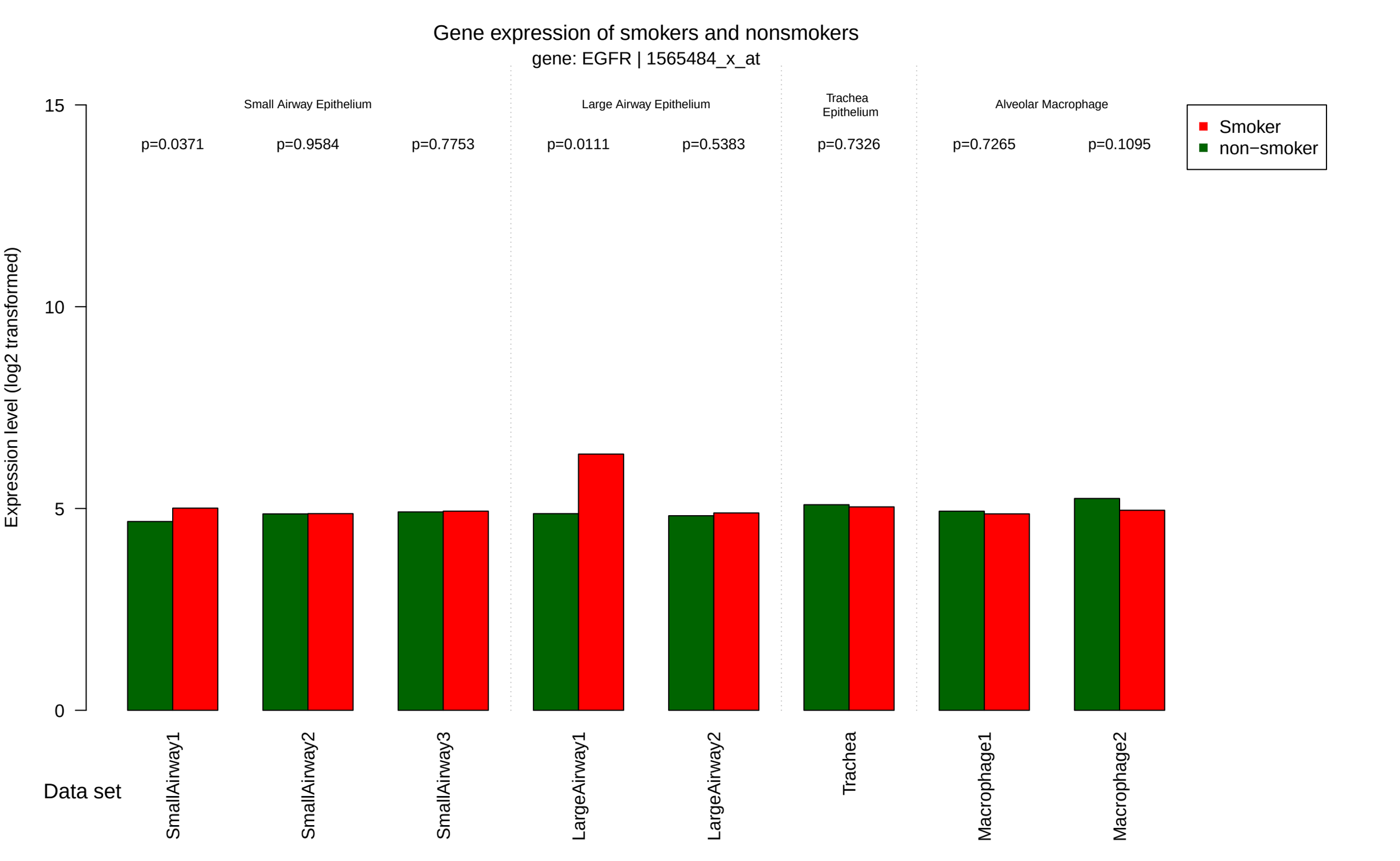
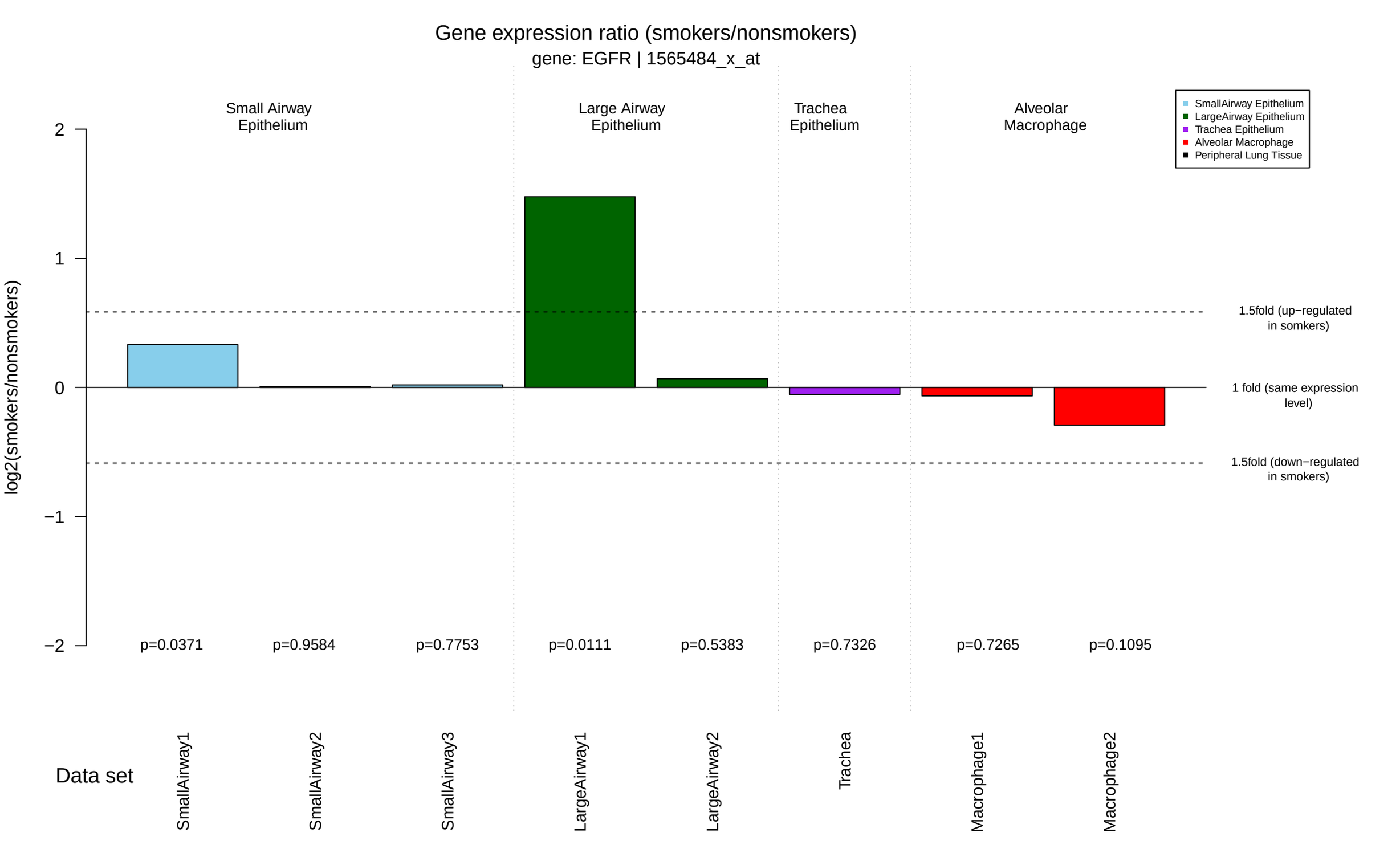
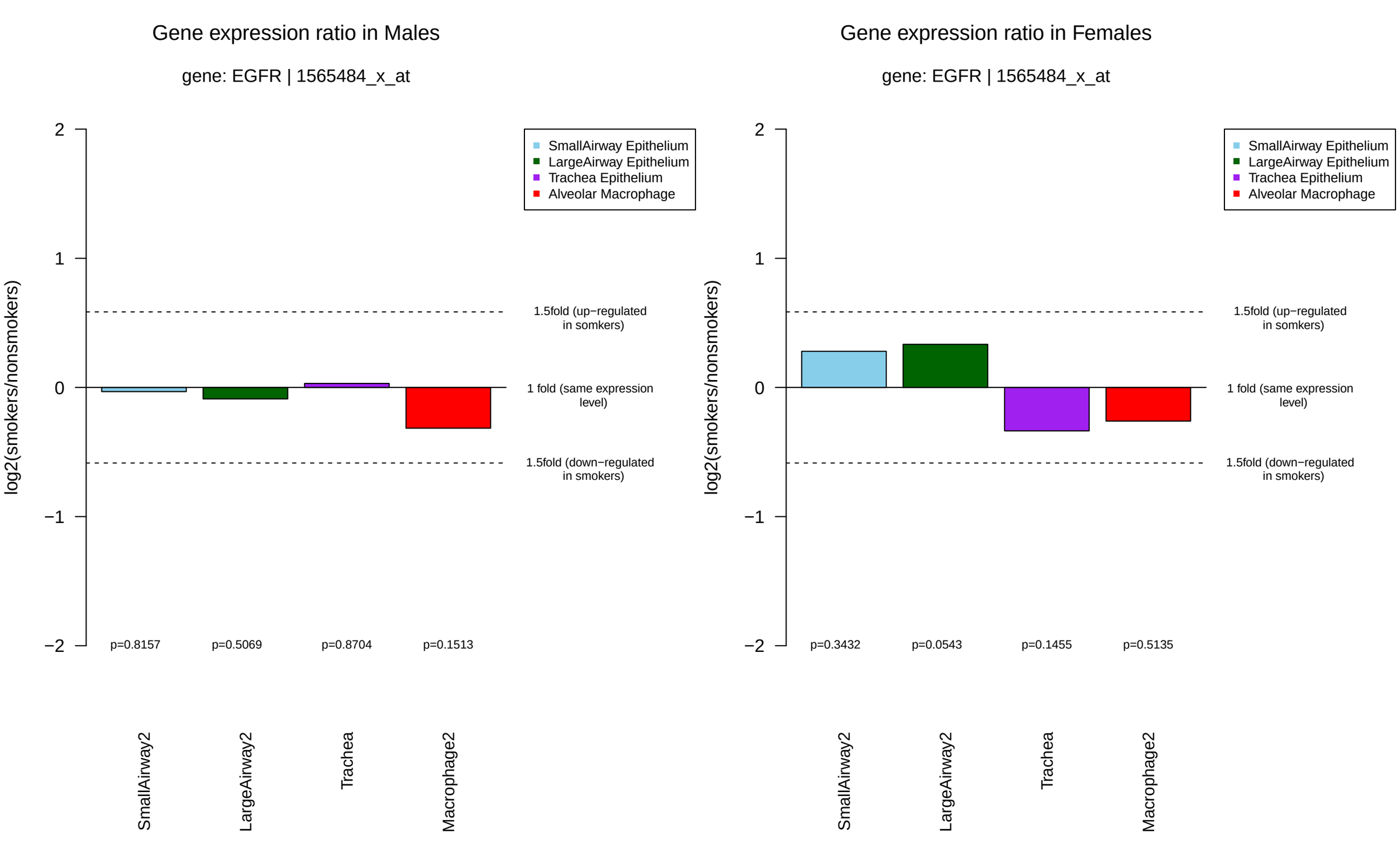
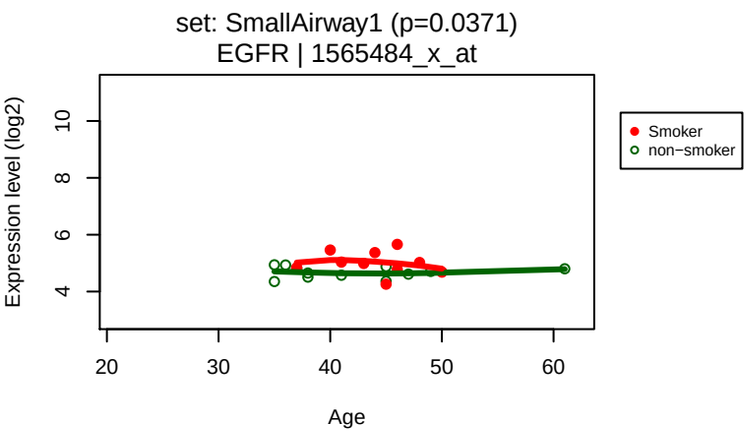
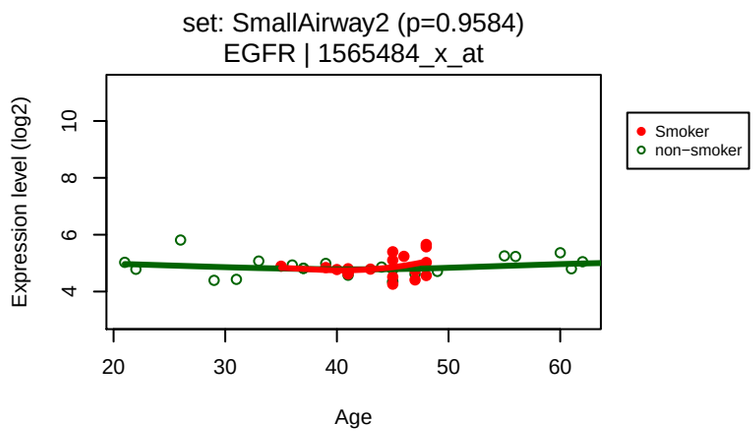
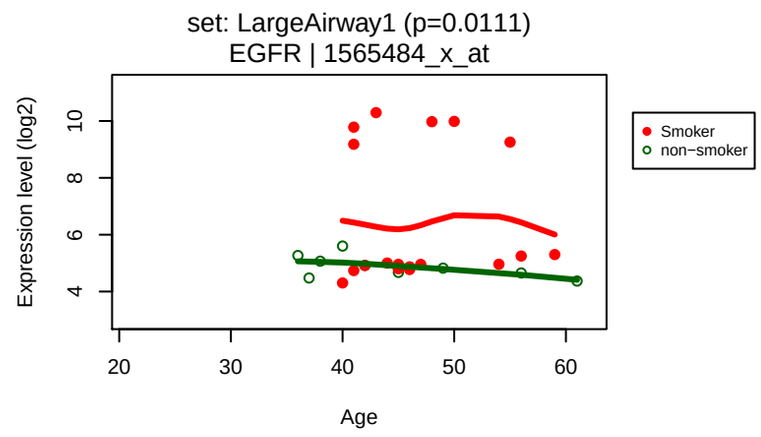

Gene expression of smokers and nonsmokers
gene: EGFR | 1565484_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0371
p=0.9584
p=0.7753
p=0.0111
p=0.5383
p=0.7326
p=0.7265
p=0.1095
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

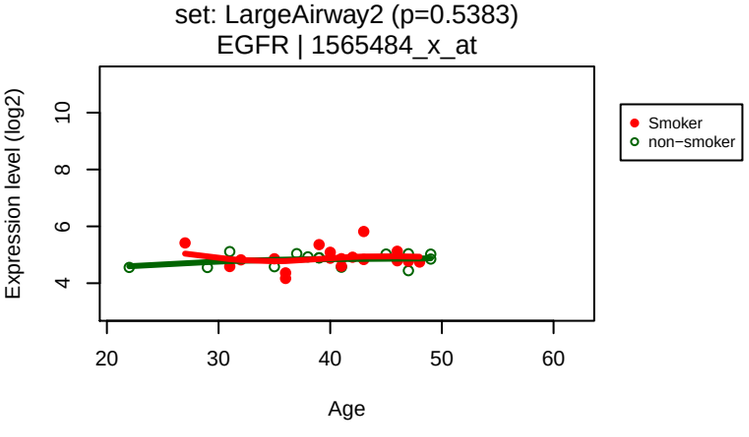

Gene expression ratio (smokers/nonsmokers)
gene: EGFR | 1565484_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0371
p=0.9584
p=0.7753
p=0.0111
p=0.5383
p=0.7326
p=0.7265
p=0.1095
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

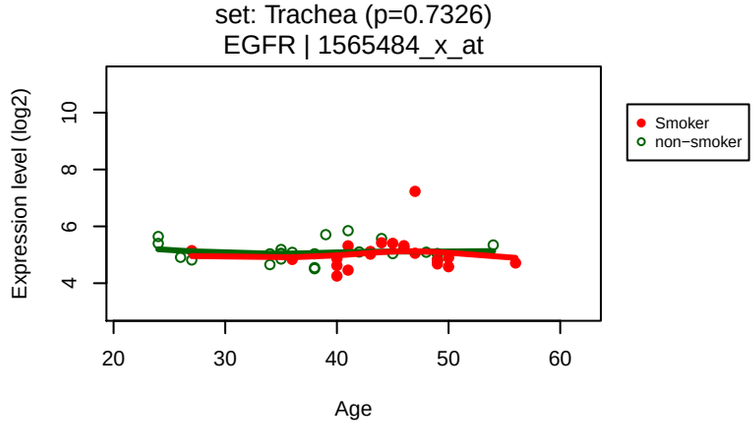

Gene expression ratio in Males
gene: EGFR | 1565484_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8157
p=0.5069
p=0.8704
p=0.1513
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: EGFR | 1565484_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3432
p=0.0543
p=0.1455
p=0.5135
SmallAirway2
LargeAirway2
Trachea
Macrophage2

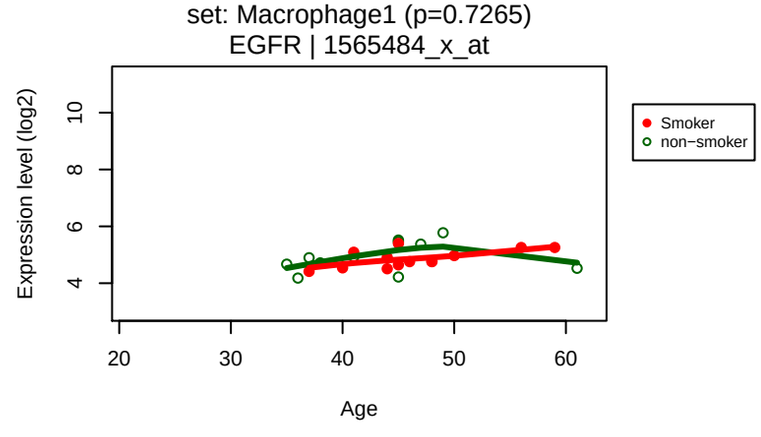

set: SmallAirway1 (p=0.0371)
EGFR | 1565484_x_at
Expression level (log2)
Age
Smoker
non−smoker

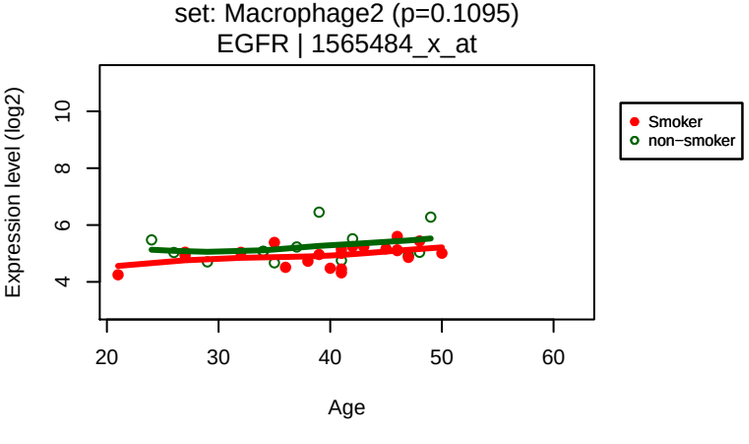

set: SmallAirway2 (p=0.9584)
EGFR | 1565484_x_at
Expression level (log2)
Age
Smoker
non−smoker

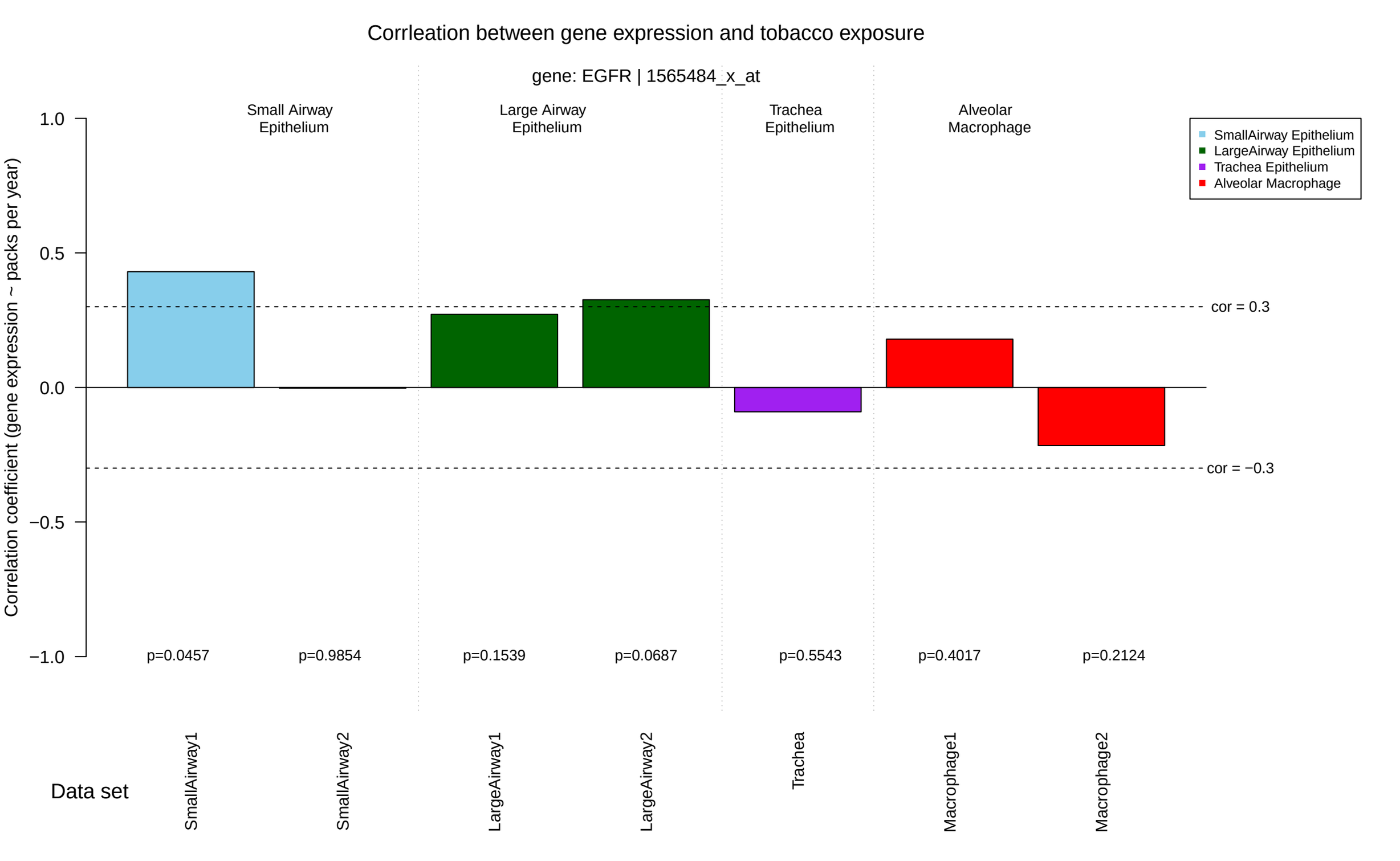

set: LargeAirway1 (p=0.0111)
EGFR | 1565484_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5383)
EGFR | 1565484_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7326)
EGFR | 1565484_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7265)
EGFR | 1565484_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1095)
EGFR | 1565484_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: EGFR | 1565484_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0457
p=0.9854
p=0.1539
p=0.0687
p=0.5543
p=0.4017
p=0.2124
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2